Real-time Embedded Object Following System Using Edge AI and FPGA-based Control

GitHub↗ 시연영상↗

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요구사항

- Raspberry Pi 5(Hailo-8), FPGA Zybo Z7-20을 활용한 **실시간 객체 추종 자율주행 전동차**
- Raspberry Pi 5(Hailo-8)을 활용한 카메라를 통한 객체 학습(Re-ID 모델 활용) 후 객체 데이터([x, dist])를 **UDP 기반 이더넷 통신**으로 Zybo Z7-20에 실시간 전송(**35ms**)
- Zybo Z7-20은 객체 데이터([x, dist])기반으로 객체 추종
- 카메라에 객체가 사라지고 3초 후, 탐색 모드로 진입(**자체 차량 회전**)
- 탐색 모드 시, **학습된 객체 이외는 인식하지 않음**
- 학습된 객체 인식 시, 다시 객체 추종 시작

담당역할

- 팀장
- 모터 드라이버 출력 Center-aligned PWM IP(PL) 설계
- Zybo Z7-20 Petalinux BSP 개발
- 객체 추종 자율주행 SW 펌웨어(PS) 개발

결과

시연평가 1등

AXI4-Lite Register Map

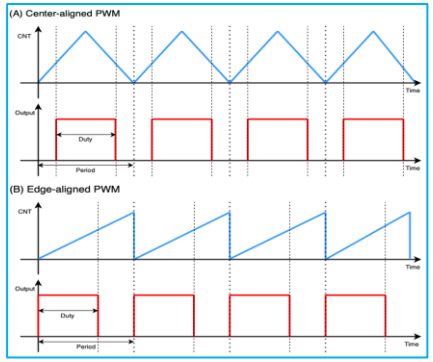
slv_reg0

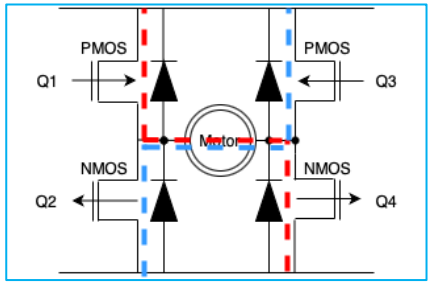
steer_stby = slv_reg0[31]	steer_dir = slv_reg0[30:29]	steer_duty = slv_reg0[28:21]	steer_time = slv_reg0[20:0]
	10 : LEFT, 01 : RIGHT		Wanted Time * 781_250 (MAX 2 Second == 2097252)

slv_reg1

drive_stby = slv_reg1[31]	drive_dir_a = slv_reg1[30:29]	drive_duty_a = slv_reg1[28:21]	no use (11 bits) = slv_reg1[20:10]	drive_duty_b = slv_reg1[9:2]	drive_dir_b = slv_reg1[1:0]
10 : FORWARD, 01 : BACKWARD				10 : FORWARD, 01 : BACKWARD	

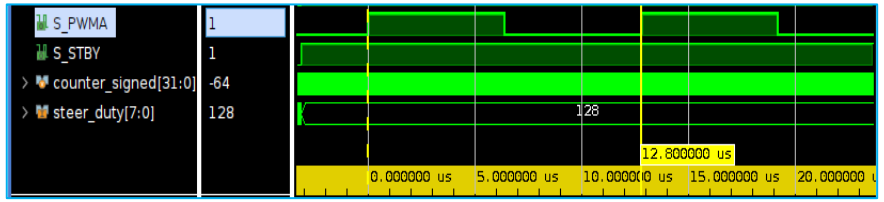
Center-aligned PWM Generator



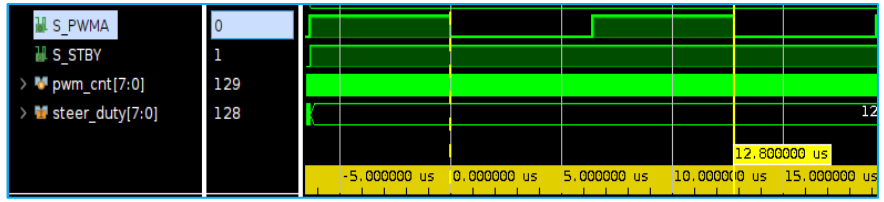


H-Bridge회로에서의 Current 흐름

Center-Aligned PWM을 통하여 Up-Down Counter로 대칭 Switching을 구현하여 **DC Motor Current Ripple을 최소화** 이로인해 H-Bridge Dead Time 관리가 용이해져 **Shoot-Through 방지 및 Switching Loss을 줄임**



(a) Center-Aligned PWM

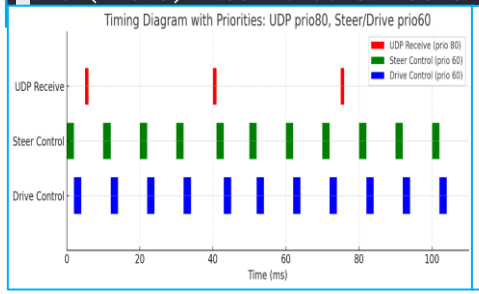


(b) Edge-Aligned PWM

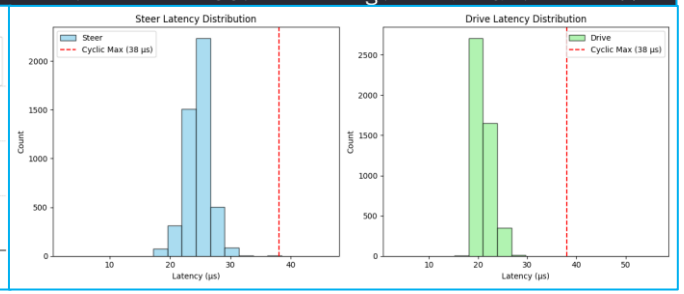
Petalinux PREEMPT RT Kernel : Benchmark

```
petalinux@capstone:~$ uname -a
Linux capstone 5.15.0-rt15-xilinx-v2022.1 #1 SMP PREEMPT_RT Sun Apr 27 07:06:49 UTC 2025 armv7l armv7l armv7l GNU/Linux
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 0.25 0.13 0.05 1/91 576

: 0 (  576) P:99 I:400 C:  95264 Min:     12 Act:    21 Avg:    17 Max:      39
```



Real-Time RT_PREEMPT Kernel



Thread Latency Distribution

Steer Control Thread Latency : **22 ~ 28 us**
Drive Control Thread Latency : **19 ~ 25 us**
< **Cyclictest Max Latency : 38 us**

Actual Processing latency : 20 ~ 30 us
Cyclictest Max Latency 보다 작은 값을 유지 (실시간성 요건 만족)